Hardware Spring Intern

Do you have a passion for computing using new age technologies?

Do you want to work on leading-edge problems alongside some of the best & brightest in the world?

Do you like working in a dynamic working environment that involves creative problem solving and thinking on your feet?

We, at NVIDIA, are in the lookout for the brightest young women technologists of our generation to do their life's best work at NVIDIA through our Flagship program #FindingHopper.

#FindingHopper, named after Grace Hopper, known for her groundbreaking work in developing computer languages, is intended for extraordinary young women to meet and learn from the industry best minds, receive networking and mentorship opportunities, professional development - all designed to inspire and set young women on a path to a promising career in tech.

We are hiring for our 6M spring internship program across multiple positions in our HW group, and the following is a brief about the teams.

GPU ARCHITECTURE

GPU architecture team is engaged in the development of industry leading high performance and power efficient GPUs. Specific areas include architecture modelling, analysis, and performance verification. The team works on GPUs across all application domains such as gaming for PC and mobile devices, professional graphics and visualization and high-performance computation.

Skills you will use/develop:

- C++ modelling, test development
- RTL design, debug
- ASIC design & verification tools, methodologies
- Computer architecture, Graphics, GPU micro-architecture, parallel computing
- Performance evaluation, analysis and debug
- Perl/Python scripting

Areas you will be working on: Computer Architecture, Memory Systems Architecture, Compiler Architecture / Performance Modelling

GPU ASIC DESIGN / VERIFICATION

Today NVIDIA's GPUs simulate human intelligence, running deep learning algorithms and acting as the brain of super computers, robots, and self-driving cars that can perceive and understand the world. We are seeking a passionate, innovative, and highly motivated senior verification engineer to join us in the development of the next generation of PCI Express controllers used in NVIDIA's GPUs and SOCs.

In this position, you will be responsible for verification of the ASIC design, architecture and microarchitecture using advanced verification methodologies. You are expected to understand the design and implementation, define the verification scope, develop the verification

infrastructure and verify the correctness of the design. You will be working with architects, designers, and pre- and post-silicon verification teams to accomplish your tasks.

What you will be doing:
- Develop test plans, tests and verification infrastructure for PCIE at IP/sub system/SOC level
- Create verification environment using UVM methodology
- Create reusable bus functional models, monitors, checkers and scoreboards
- Drive functional coverage driven verification closure
- Work with architects, designers and post silicon teams

Ways to stand out from the crowd:
- Good knowledge of PCIE protocol Gen 3 and above
- Good debugging and problem-solving skills
- Good communication skills
- Ability/ desire to work as a team player

TEGRA SOC DESIGN / VERIFICATION

As part of this team, you will design and implement the industry's leading Graphics, Video and Mobile Communications Processors. Specific areas include 2D and 3D graphics, mpeg, video, audio, network protocols, high-speed IO interfaces and bus protocols, and memory subsystem design. You will be responsible for Architecture and microarchitecture design of the ASICs, RTL design and synthesis, Logic and Timing verification using leading edge CAD tools and Semiconductor process technologies.

Areas you will be working in:
- ASIC, RTL, Design and Verification of Processors
- Low Power verification
- Power Estimation and Modelling
- PCIe Design Verification
- Functional / Formal verification

CPU ARCHITECTURE / DESIGN/ VERIFICATION

As a design and verification/validation engineer in the ARM CPU team, you will be working on the next generation of 64bit ARM CPUs and SOCs. As part of this assignment, you will get a chance to learn about computer architecture at a very granular level, System Verilog, Design Verification, SOC Verification, Verification methodologies and C/C++ programming. You also will get an opportunity to get familiar with industry standard tools in verification and validation.

During the course of the internship, you will contribute to building test benches, developing architectural simulators, modifying random instruction generators and creating stimulus for verification and validation of different units of the CPU and SOC.

Areas you will be working in:
- Computer Architecture
- Digital Design and Programming in C/C++/Perl
- ARM, CPU Design and Verification/ Validation

VLSI

VLSI team works in the areas of RTL Design, Verification, Design for Test, CAD tools, Synthesis, Timing, Place & Route, Circuit Design, Silicon Bring-up and Characterization. Responsible for state-of-the-art methodologies, tools, flows and project execution on all Nvidia GPU, CPU, Auto, and Switch chips. As an intern you will be working on one or more such areas.

Skills you will use/develop:
- RTL Design, VCS, SV, UVM, Formal
- Verilog, C/C++, Python, TCL, Perl
- Logic Scan Test, Memory Test, High-speed IO Test, In-System Test
- Synthesis, Timing Closure (Primetime)
- Physical Design, Innovus, ICC2, Physical extraction, Place and Route, Floorplan
- SRAM, Analog, Digital circuit design, Hspice, EMIR, Silicon Correlation (ATPG, data visualization & analysis)
- Layout (Cadence Virtuoso)
- Silicon characterization for ageing, DPPM, Signal Integrity, Power Integrity, and familiarity with PC/SOC sub-system Architecture

COMPUTE DEVELOPER TECHNOLOGY (DevTech)

Our team works with cutting-edge AI workloads and NVIDIA GPUs to solve difficult performance problems. We analyze applications, develop optimized implementations, and collaborate with teams across NVIDIA to improve the hardware and software platform.

This internship is ideal for students who enjoy understanding how computers work beneath high-level frameworks. We are particularly interested in candidates who have knowledge of hardware architecture, parallel programming, and performance optimization and who want to learn more. An interest in system architecture and end-to-end system behavior is a plus.

What you will be doing:
- Develop and optimize GPU-accelerated techniques for deep learning, graph analytics, machine learning, and data-intensive applications.
- Analyze workloads to identify bottlenecks in computation, memory access, synchronization, and data movement.
- Implement and evaluate parallel algorithms and CUDA C/C++ kernels on current GPU architectures.
- Use profiling and benchmarking tools to measure performance, validate improvements, and explain results.

Required Skills: Programming/C/C++, DS, Algo, OS, Comp Arch, Compilers, high-performance computing, or ML systems

COMPUTE ARCHITECTURE

As part of our Deep Learning Architecture team, you can help build real-time, cost-effective computing platforms driving our success in this exciting and rapidly growing field.

What you will be doing:
- Develop innovative HW architectures to extend the state of the art in parallel computing performance and energy efficiency.
- Benchmark and analyze AI workloads in single and multi-node configurations.
- Develop tools to profile, analyze and debug parallel applications in Python/C++.
- Work closely with peer architecture teams and product management to guide development of the products.
- Keep abreast with emerging trends and research in deep learning.

Required Skills: C, C++ and Python, excellent problem solving, understanding of deep learning and AI workloads, familiarity with GPU computing and parallel programming, experience with architecture simulator development, performance modelling, benchmarking, profiling, analysis and debugging, and strong collaboration skills